

Interactions in quantum networks with pulse propagation delays

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The finite speed of light of pulses implies propagation delays between the generation of light, the interaction with different elements along its path, and the final detection at the output of networks and interferometers. We show that it is possible to take these delays into account with a theoretical method which evades quantization of the full continuum of field modes. We illustrate the use of this method by analyzing Ramsey excitation of an atom by a split and delayed quantum pulse.

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I. INTRODUCTION

Quantum networks are spatial architectures where propagating quantum fields enable the transfer of states, excitation, or entanglement between different nodes. Such networks may display novel entanglement properties [1] and they may serve practical purposes such as clocks [2], sensors [3,4] and forming the basis of secure quantum communication [5] and distributed optical quantum computing [6–8].

The fact that propagating quantum fields occupy a continuum of modes, e.g., wave-number eigenmodes or coupled space-bin modes, poses a challenge to their theoretical description. This challenge can be alleviated when propagation times are much shorter than the natural timescale of physical interactions, in which the interaction with multiple components can be treated as simultaneous. If an arbitrarily extended network permits only unidirectional and nondispersive propagation, one can absorb propagation delays in the synchronization of clocks at each node such that, also in this case, the interactions appear formally simultaneous. It is subsequently possible to employ the Born-Markov approximation to eliminate the field modes altogether and obtain coupled master equations of the systems located at the network nodes.

Such equations were introduced by Gardiner and Collett [9,10] and Carmichael [11] and were referred to as cascaded master equations to reflect the fact that the output from one node serves as input to other ones in the network. A general so-called SLH formalism, named after matrix components S , L , and H , provides an efficient determination of the master equation for general architectures of concatenated network elements [12,13].

When the assumption of negligible time delay between the interaction with different components breaks down or cannot be remedied by a local synchronization of time, such as in the example network in Fig. 1 where delay lines are an integral

part of the network dynamics, the efficient cascaded master equation does not apply and more complex methods are needed. We note in particular the quantization of the propagating field across discrete bins in space or time, which explores a Hilbert space of vast dimension but does permit an effective treatment by matrix product states. This approach has been implemented to account for propagation and data processing delays in the interesting case where measurements of output signals are used to guide feedback operations on the emitters (delayed with respect to the time of emission) [14–17]. Another approach, which is more similar to the approach we take here, is to introduce fictitious elements that model the effect of time delays to some perturbative order [18,19].

In many applications of quantum networks, the focus is on the quantum states held by the nodes and carried by propagating wave packets of radiation. A wave packet is a single mode and its quantum state is analogous to the states of a single-mode cavity field, for which interactions with atomic systems have been studied in great detail. An important difference between cavity eigenmodes and propagating fields, however, is that the former have discrete frequencies and only

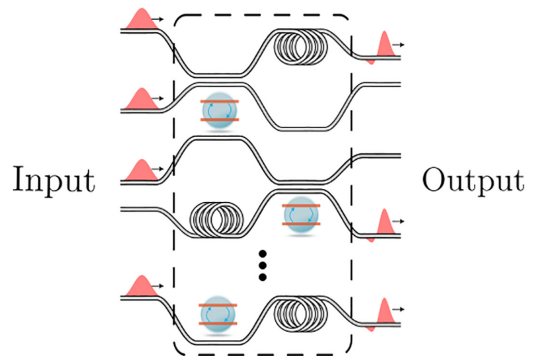


FIG. 1. Example of a quantum network with several input and output quantum channels with integrated delay lines and nonlinear components (e.g., two-level systems) illuminated by pulses of quantum radiation.

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one mode may couple resonantly to a given emitter, while propagating fields explore a mode continuum. Wave packets are hence subject to both dispersion and entanglement between the number of quanta and their spatiotemporal profile by nonlinear scatterers. The latter may be seen as a loss of (single-mode) fidelity, but also as a resource for genuine multimode functionalities [20,21]. Formally, the joint evolution of the photon-number statistics and propagation of the wave packet can be taken into account by an effective formalism that represents input traveling pulses as the output from virtual cavities and subsequently determines the multimode character of the output fields [22–25].

In this article we are interested in how propagation delays affect the interaction of pulses of light with scattering elements. With a focus on the physics of single-mode pulses exploring unidirectional waveguides, we propose to represent the time delays by effectively capturing a single or few wave packets in virtual cavities and releasing them with the desired delay. Unlike the discretization in time bin modes, we hence only need one single auxiliary mode to describe the time delay of a pulse, irrespective of the duration of the delay, while enjoying a nonperturbative approach in contrast to the method of using fictitious elements.

The article is structured as follows. In Sec. II we present the virtual cavity method for delaying pulses of quantum radiation in a network. In Sec. III we discuss a simple example of how to use the method to delay the field in one channel in a network with respect to another one and we present a method to reduce the complexity that arises from using the method presented in Sec. II. In Sec. IV we employ our method to analyze illumination of an atom by a radiation pulse that has been split at a beam splitter and its components have been delayed with respect to each other. This is the central process in Ramsey spectroscopy, where the use of classical, coherent pulses of radiation are treated by a simple atomic master equation. We generalize this treatment to illumination by entangled pulses of quantum radiation. In Sec. V we analyze the intensity of the output radiation from the atom. In Sec. VI we summarize and indicate applications and natural next directions of research building on the pulse quantization and virtual cavity approach.

II. THEORETICAL MODEL FOR DELAYED PULSES OF RADIATION

In experiments, pulses of light are delayed if they are sent along extended optical paths. This article is not about experimental schemes to generate time delays, but about the theoretical description of interactions between quantum systems and pulses of light in networks with inherent time delays. Rather than employing a quantized version of Maxwell's equations for the propagating field, the delay of the time of arrival of a quantum pulse due to a propagation distance can be simulated by the capture and subsequent release of the pulse. For our purposes, we will simulate pulse delays by the equivalent process of storing the pulse in a cavity mode and releasing it at a later time.

In [22–25] virtual cavities were similarly used to model the launching of a quantum pulse with pulse shape $u(t)$ onto any downstream system, by assuming the initial quantum state of the pulse is prepared in an eigenmode of a cavity, which is

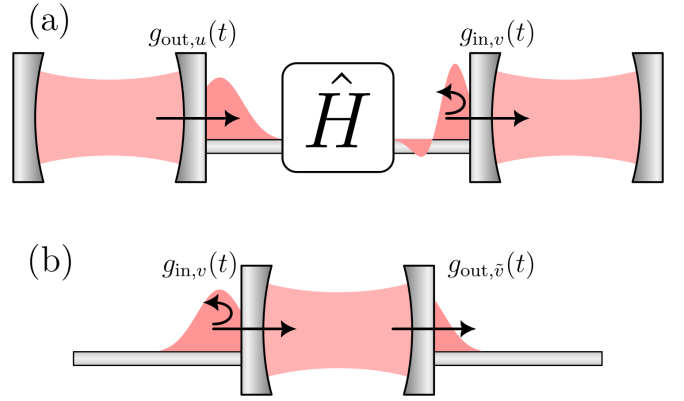


FIG. 2. Virtual cavity approach to pulses of quantum radiation. (a) A pulse mode $u(t)$ in any desired quantum state can be theoretically modeled as the output from a single-mode cavity. The interaction with a quantum node and the quantum state contents of any downstream wave packet $v(t)$ can be obtained by solving a cascaded master equation for the emitting u cavity, the quantum node, and an appropriate v cavity. (b) A pulse $v(t)$ can be delayed by first storing it in a cavity mode and subsequently releasing it in mode $\tilde{v}(t) = v(t - \tau)$, described formally by the same process as shown in (a). Pulses orthogonal to the chosen one will be reflected by the delay cavity and will not be delayed, unless other designated delay cavities are employed to that effect.

coupled to the environment with the time-dependent coupling strength,

$$g_{out,u}(t) = \frac{u^*(t)}{\sqrt{1 - \int_0^t dt' |u(t')|^2}}. \quad (1)$$

Similarly, to perfectly transfer radiation occupying a pulse with the shape $u(t)$ into a cavity eigenmode, the coupling to this cavity must be

$$g_{in,u}(t) = \frac{-u^*(t)}{\sqrt{\int_0^t dt' |u(t')|^2}}. \quad (2)$$

We will employ the same formalism to model the delay of a quantum pulse by its absorption in a virtual cavity and its reemission at a chosen later time, as pictured in Fig. 2. The pulse is first given by $v(t)$ and is absorbed by the virtual cavity with coupling rate $g_{in,v}(t)$. Hereafter, the cavity coupling rate of the output port follows $g_{out,u}(t)$ to emit the pulse in the desired shape $u(t)$. If only a delay is desired, we choose $u(t) = v(t - \tau)$, where τ is the desired time delay. Note that this stepwise approach functions only if τ is large enough that the pulse can be fully captured before it is rereleased, where fully captured means all but a negligible tail of the pulse has been captured. In the following sections, we will discuss modifications of the theoretical scheme that permit treatment of arbitrarily short delays.

In the same way, several simultaneously occupied pulses of radiation can all be delayed by storage in and release from a cascaded array of virtual cavities with appropriate time-dependent coupling strengths [22,23]. It may seem that employing an auxiliary oscillator mode for the delay of each pulse will extensively complicate the calculation of the

dynamics. However, we will show in the following that since the delaying dynamics is linear, an effective interaction picture applies and simplifies the calculations considerably [24,25].

A. Delaying a single-mode field by an arbitrary amount

Fully absorbing and then reemitting pulses from a virtual cavity is incompatible with delays τ shorter than the pulse duration, since the release out of the cavity would happen before the pulse is assumed to be fully absorbed. For such short delays, we find a time-dependent cavity coupling that ensures the desired balance between the input and output amplitudes at any time while storing part of the field inside the cavity for later release. We note that in Eqs. (1) and (2), the denominator represents the fraction of the pulse residing inside the emitting and absorbing cavities, respectively. To account for the fact that the cavity is fed by an incoming pulse $v(t)$ while at the same time radiating an outgoing pulse $u(t)$, we simply change 1 in the denominator of Eq. (1) to the integrated radiation that has entered the cavity so far. This means we get the out-coupling strength

$$\tilde{g}_{\text{out},u,v}(t) = \frac{u^*(t)}{\sqrt{\int_0^t dt' |v(t')|^2 - \int_0^t dt' |u(t')|^2}}, \quad (3)$$

where $v(t)$ is the shape of the arriving pulse being gradually absorbed while the cavity also leaks a pulse in the shape of $u(t)$. At the same time, modifying the assumption that the cavity does not lose any radiation while it picks up mode $v(t)$ implies that the incoupling strength, through the input mirror, takes the value [cf. Eq. (2)]

$$\tilde{g}_{\text{in},v,u}(t) = \frac{-v^*(t)}{\sqrt{\int_0^t dt' |v(t')|^2 - \int_0^t dt' |u(t')|^2}}. \quad (4)$$

With these expressions, we can convert radiation between pulse shapes as long as the integrated emitted power is at any time less than or equal to the integrated incident power. We are interested in the simple delay $u(t) = v(t - \tau)$, and we observe that when τ is larger than the duration of $v(t)$, Eqs. (3) and (4) take the form of Eqs. (1) and (2) in two consecutive time intervals.

Equations (1)–(4) apply for any (square-integrable) pulse, including mathematical pulse shapes, e.g., Gaussians, that only vanish asymptotically far before and after the pulse peak. To be ensured that we account for a fraction $1 - \varepsilon$ arbitrarily close to unity of the entire input state, we pick a sufficiently big cutoff for the finite-time integration around the pulse peak at t_p , $\int_{t_p-2t_c}^{t_p+2t_c} |u(t)|^2 dt = 1 - \varepsilon$, where we denote by $2t_c$ the pulse length. In practice, a t_c of a few times the width t_w of the Gaussian pulse will suffice and a similar identification of a proper cutoff time for the pulse length can be made for any square-integrable pulse shape.

B. Delaying a multimode field

We have not found a way to use the same method and a single cavity to delay fields containing multiple pulse modes; instead, we have recourse to sequentially fully absorb all relevant input pulses $v_i(t)$ in separate cavities,

before reemitting them a time $\tau + T$ later, where τ is an adjustable relative delay and T is the time it takes to fully absorb the pulses, which for smoothly tapered pulses will imply a small but negligible error associated with the tail of the pulses.

In this case, it is not possible to delay the pulses by an arbitrarily small amount, as the smallest delay possible is T . However, in most situations, it is not the actual delay that is important, but rather the relative delay between different propagation channels, as in an interferometer. In such cases, it is not necessary for the physical pulse duration to be shorter than the delay time. Capturing all pulses and sending them with a relative delay reproduces the physical functioning of the original setup if we change the time argument by that same offset amount when reporting the prediction for the experiment. In the next section we will show this for a Mach-Zehnder interferometer with different path lengths. For simplicity, we consider delaying only a single pulse, but the method may be extended to a multimode field by including several delay cavities.

III. MACH-ZEHNDER INTERFEROMETER WITH DIFFERENT PATH LENGTHS

In Mach-Zehnder interferometers, one optical path is longer than the other, and an incident pulse may be split and the components may be delayed with respect to each other. In the continuous-wave regime, this would be described by a phase shift of the radiation in one path, while for pulses of finite-time length, the effect is more complicated.

To simulate the relative delay between two paths, we use the method described in the preceding section, assuming virtual delay cavities along both paths to pick up the radiation that is scattered on the beam splitter. Once the radiation has been picked up in both paths at time T , the radiation is reemitted first from the delay cavity representing the shorter path and, after a relative delay τ , by the cavity representing the longer path. With this method, we can model any path length difference. Note that while pulse scattering on a linear beam splitter may be resolved by other means, our method is more general and also applies to scattering off nonlinear systems. We give an example of this in Sec. V.

It seems to be a significant disadvantage of our method that it generally needs two oscillators to delay one pulse relative to another one, and the Hilbert space dimensionality has hence increased drastically to account for the delay. Since, however, the input field is single mode and the beam splitter merely performs a mode transformation, the system remains single mode in the proper basis. Furthermore, we can simplify the linear dynamics by going to an appropriate interaction picture.

We consider a setup where a virtual cavity is used to describe the pulse inbound on a beam splitter with pulse shape $u(t)$. To simulate the delay between the paths of the interferometer, a delay cavity is placed in each path designed to pick up the radiation in the same mode $u(t)$ and reemit it at a later time in modes $v_1(t) = u(t - \tau - T)$ and $v_2(t) = u(t - T)$, where T is chosen such that both cavities have picked up the entire incoming $u(t)$ mode and τ is the relative delay of the long path compared to the short path. The Hamiltonian for

this interaction is

$$\hat{H} = \frac{i}{2\sqrt{2}} [g_{\text{out},u}(t)g_{\text{in},u}^*(t)\hat{a}_u^\dagger\hat{a}_{v_1} + g_{\text{out},u}(t)g_{\text{in},u}^*(t)\hat{a}_u^\dagger\hat{a}_{v_2} - \text{H.c.}], \quad (5)$$

where $\hat{a}_u$ is the annihilation operator for the virtual input cavity, $\hat{a}_{v_1}$ and $\hat{a}_{v_2}$ are the annihilation operators for the delay cavities representing the long and short paths, respectively, and H.c. denotes the Hermitian conjugate. The Hamiltonian evolution is accompanied by two dissipation operators for the two output channels, which reflect the input field

$$\hat{L}_{r1} = \frac{1}{\sqrt{2}}g_{\text{out},u}^*(t)\hat{a}_u + g_{\text{in},u}^*(t)\hat{a}_{v_1}, \quad (6)$$

$$\hat{L}_{r2} = \frac{1}{\sqrt{2}}g_{\text{out},u}^*(t)\hat{a}_u + g_{\text{in},u}^*(t)\hat{a}_{v_2}, \quad (7)$$

and further two dissipation operators that carry the delayed field

$$\hat{L}_{t1} = g_{\text{out},v_1}^*(t)\hat{a}_{v_1}, \quad (8)$$

$$\hat{L}_{t2} = g_{\text{out},v_2}^*(t)\hat{a}_{v_2}. \quad (9)$$

We go to the interaction picture with respect to the Hamiltonian, which describes the transfer of excitation from the input cavity to the delay cavity before the pulse is rereleased. We define the operators $\hat{b}_1 = (\hat{a}_{v_1} + \hat{a}_{v_2})/\sqrt{2}$ and $\hat{b}_2 = (\hat{a}_{v_1} - \hat{a}_{v_2})/\sqrt{2}$. In the interaction picture, the time evolutions of the operators are

$$\frac{d}{dt}\hat{a}_u = \frac{1}{2}g_{\text{out},u}(t)g_{\text{in},u}^*(t)\hat{b}_1, \quad (10)$$

$$\frac{d}{dt}\hat{b}_1 = -\frac{1}{2}g_{\text{out},u}^*(t)g_{\text{in},u}(t)\hat{a}_u, \quad (11)$$

$$\frac{d}{dt}\hat{b}_2 = 0. \quad (12)$$

The solution to these equations is found in [24,25] for equal input and output pulses and is given by

$$\hat{a}_u(t) = \cos\theta(t)\hat{a}_u(0) - \sin\theta(t)\hat{b}_1(0), \quad (13)$$

$$\hat{b}_1(t) = \cos\theta(t)\hat{b}_1(0) + \sin\theta(t)\hat{a}_u(0), \quad (14)$$

where $\theta(t)$ is defined by the relation

$$\sin^2\theta(t) = \int_0^t dt' |u(t')|^2. \quad (15)$$

The Hamiltonian in the interaction picture thus vanishes, $\hat{H}_I = 0$, as the interaction picture incorporates the full Hamiltonian dynamics. The dissipation operators simplify

$$\hat{L}_{r1,I} = \frac{1}{\sqrt{2}}[-2u(t)\csc(2\theta)\hat{b}_1 + g_{\text{in},u}^*(t)\hat{b}_2], \quad (16)$$

$$\hat{L}_{r2,I} = \frac{1}{\sqrt{2}}[-2u(t)\csc(2\theta)\hat{b}_1 - g_{\text{in},u}^*(t)\hat{b}_2] \quad (17)$$

and

$$\hat{L}_{t1,I} = \frac{1}{\sqrt{2}}g_{\text{out},v_1}^*(t)(\cos\theta\hat{b}_1 + \sin\theta\hat{a}_u + \hat{b}_2), \quad (18)$$

$$\hat{L}_{t2,I} = \frac{1}{\sqrt{2}}g_{\text{out},v_2}^*(t)(\cos\theta\hat{b}_1 + \sin\theta\hat{a}_u - \hat{b}_2). \quad (19)$$

As the incoupling to the delay cavities is designed to fully absorb the incoming pulse shape and reflect only vacuum, we can omit terms with $\hat{L}_{r1,I}$ and $\hat{L}_{r2,I}$ from the master equation, as they will have no effect, since no radiation leaves along these channels. Furthermore, since the emission of the delayed modes governed by $g_{\text{out},v_i}(t)$ happens after the loading of the pulse into the delay cavities governed by $u(t)$, we can replace the trigonometric functions by their long-time limits $\cos\theta \rightarrow 0$ and $\sin\theta \rightarrow 1$ in $\hat{L}_{t1,I}$ and $\hat{L}_{t2,I}$. We observe then that the $\hat{b}_1$ operator disappears in this interaction picture. The dynamics thus only introduces one additional oscillator mode to account for the relative delay of the incoming pulses. In the next section we will release the two pulses onto a two-level emitter and thereby investigate a Ramsey-type interaction of quantum pulses of light with an atom.

We end the section by noting how to incorporate transmission loss in the formalism, which may be different for each quantum channel due to the difference in length. We model this loss through reflection and transmission through a beam splitter, where the reflection coefficient simulates the loss, $\sqrt{R_i}$, where R_i is the transmission loss of the i th channel, which may scale with length. The beam splitter introduces another dissipation channel for the transmission loss, and the dissipation operator $\hat{L}_i$ for the i th channel in the network maps onto the loss channel as $\sqrt{R_i}\hat{L}_i$ and the further propagating channel as $\sqrt{1-R_i}\hat{L}_i$. For the above case for the Mach-Zehnder interferometer, the propagating channels are

$$\hat{L}_{p1} = \sqrt{1-R_1}\hat{L}_{t1,I}, \quad (20)$$

$$\hat{L}_{p2} = \sqrt{1-R_2}\hat{L}_{t2,I}, \quad (21)$$

while the transmission loss channels are

$$\hat{L}_{l1} = \sqrt{R_1}\hat{L}_{t1,I}, \quad (22)$$

$$\hat{L}_{l2} = \sqrt{R_2}\hat{L}_{t2,I}. \quad (23)$$

In the following section we neglect this loss ($R_i = 0$), but the loss can be readily incorporated by using the propagation dissipation operators $\hat{L}_{pi}$ instead of $\hat{L}_{ti,I}$ in the following derivations and including $\hat{L}_{li}$ as dissipation operators in the master equation.

IV. EXPOSING A NONLINEAR SCATTERER TO DELAYED QUANTUM PULSES

So far, our analysis and methods have concerned only linear propagation and mode mixing, which may be readily solved in the Heisenberg picture. In this section we consider a nonlinear scattering process with the setup sketched in Fig. 3(a), where a two-level system is illuminated by two pulse components with a time delay between them. This constitutes a quantum version of Ramsey spectroscopy, and we aim to show the effect of coherences between the pulses on the atom when the light pulses have no mean-field amplitude but are entangled with each other.

The Hamiltonian for the interaction is

$$\hat{H} = \Delta\hat{\sigma}_+\hat{\sigma}_- + \frac{i}{2}\sqrt{\gamma}\left[g_{\text{out},v_1}(t)\hat{c}_+^\dagger\hat{\sigma}_- + g_{\text{out},v_2}(t)\hat{c}_+^\dagger\hat{\sigma}_- - \text{H.c.}\right], \quad (24)$$

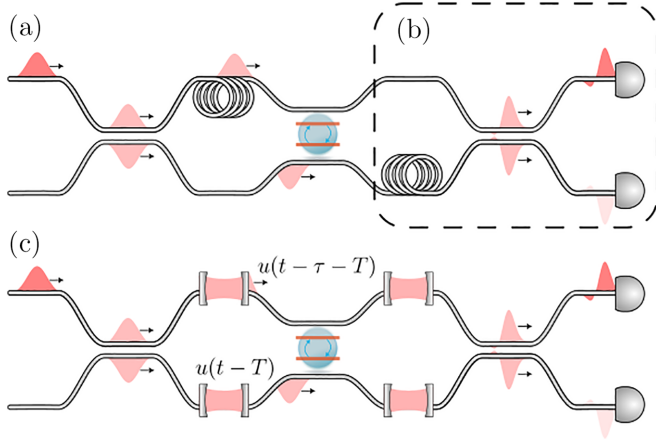


FIG. 3. Ramsey-like interferometry experiment. (a) A pulse of quantum radiation is split on a beam splitter before entering a long and a short path. Both paths pass in the vicinity of a two-level system, which interacts with the pulses of quantum radiation. (b) The outputs after interaction with the two-level system are further delayed to be superimposed after the same total delay on a beam splitter and subsequently detected by photodetectors placed in both paths. (c) The difference in path length is modeled by virtual delay cavities that emit the pulses with a relative delay τ to emulate the effect of the different path lengths in the physical network.

where $\hat{\sigma}_{\pm}$ are operators associated with the creation or destruction, respectively, of a quantum of excitation in the two-level system, Δ is the detuning between the incoming radiation and the energy levels of the two-level system, and we choose the two-level atom to have a decay rate $\gamma/2$ along each quantum channel, resulting in a total decay rate γ . We have furthermore introduced the operators $\hat{c}_{\pm} = (\hat{a}_u \pm \hat{b}_2)/\sqrt{2}$ to ease the notation. The Hamiltonian is accompanied by the two dissipation operators

$$\hat{L}_1 = g_{\text{out},v_1}(t)\hat{c}_+ + \sqrt{\frac{\gamma}{2}}\hat{\sigma}_-, \quad (25)$$

$$\hat{L}_2 = g_{\text{out},v_2}(t)\hat{c}_- + \sqrt{\frac{\gamma}{2}}\hat{\sigma}_-. \quad (26)$$

We now turn to integrating the corresponding master equation

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}(t), \hat{\rho}] + \sum_{i=1}^2 D[\hat{L}_i(t)]\hat{\rho}, \quad (27)$$

where $D[\hat{L}]\hat{\rho} \equiv -\frac{1}{2}(\hat{L}^\dagger\hat{L}\hat{\rho} + \hat{\rho}\hat{L}^\dagger\hat{L}) + \hat{L}\hat{\rho}\hat{L}^\dagger$. We integrate the system numerically using the qutip toolbox [26–28]. To investigate if Ramsey-like interference occurs when the atom interacts with a Fock state split into an early and a late pulse, we vary the detuning Δ while keeping the relative delay τ between the pulses constant. We assume an input pulse with a Gaussian temporal profile given by

$$u(t) = \frac{1}{\sqrt{t_w}\sqrt{\pi}} \exp\left(-\frac{(t-t_p)^2}{2t_w^2}\right), \quad (28)$$

where t_p is the time at which the pulse peaks and t_w is the width of the pulse. To explore different regimes of photon

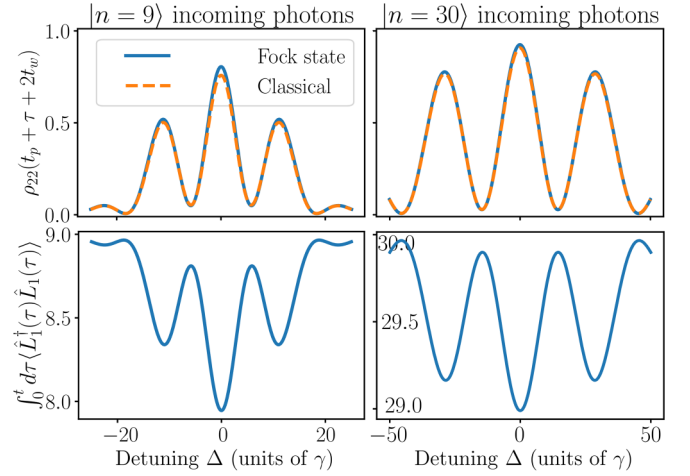


FIG. 4. The blue solid line in the two top figures show the average atomic population as a function of detuning at two pulse widths after the peak of the second pulse at time $t = t_p + \tau + 2t_w$ for driving with a split-delayed Fock state as described in the text. The population clearly show a Ramsey-like behavior for both the $n = 9$ and the $n = 30$ incoming photonic Fock state. The orange dashed line in the same figures shows the atomic population driven by two equivalent classical pulses, where each pulse has an average photon number equal to $\alpha = \sqrt{n/2}$. The two setups show the same qualitative behavior. The bottom panels show the integrated output intensity of the light in the initial mode $u(t)$ up to time $t = t_p + \tau + 2t_w$ in the output port where the photons would interfere constructively if there was no atom. The intensity clearly shows a similar behavior as the atomic population, meaning that the population of the atom on average decays into orthogonal modes to the incoming mode. We make it clear that this does not mean the output Fock state is $|n-1\rangle$, only that the average number of photons approaches $n-1$ in the zero-detuning case.

numbers, we investigate two cases with $n = 9$ and $n = 30$. For the Ramsey sequence, we choose the relative delay between the pulses to be $\tau = 0.5\gamma^{-1}$ for $n = 9$ and $\tau = 0.2\gamma^{-1}$ for $n = 30$ and the width of the pulses $t_w = \pi^{3/2}/8\gamma n$, which is the pulse length that gives a $\pi/2$ rotation of the two-level system by a field with mean photon number n in the classical limit.

Figure 4 shows the atomic population as a function of detuning. Since the atom experiences decay on the same timescale as the pulse width, we plot the excited-state population at a time equal to two pulse widths after the peak of the second pulse $t_1 = t_p + \tau + 2t_w$, where the atom has not fully decayed yet. The time t_1 is chosen after the interaction with both pulses is almost completely over and before the subsequent spontaneous decay of the atom to optimally show the Ramsey interference pattern in the atomic excited-state population in Fig. 4. Our numerical solution of the cascaded master equation may be used to further optimize the value of t_1 . The same intuitive procedure applies also for non-Gaussian pulses. While t_w is not defined in such cases, one might relate the chosen value of t_1 to another similar parameter for the given pulse shape. In Fig. 4 we notice that the atomic population experiences a similar behavior as in the classical Ramsey effect, where the detuning dependence of the

classically driven excited-state population in the final state comes from the interference between two excitation pathways $|g\rangle \rightarrow |g\rangle + e^{i\phi(t)}|e\rangle \rightarrow |e\rangle$. In the quantum case the drive is not classical, but instead the atom is driven by a Fock state $|n\rangle$, which exhibits no classical coherence. Here the binomial splitting of the Fock state on the beam splitter leads to a coherent superposition of Fock product state components. It is then the interference between pathways that have exchanged and not exchanged a photon between the early and late pulses via the atom that results in the Ramsey-like interference pattern. To clarify this similarity, we show in Fig. 4 the atomic state population driven by two classical pulses with an average photon number $\alpha = \sqrt{n/2}$, but otherwise with the same parameters as the quantum pulse (see Appendix B). We stress that although the results are similar, the physical mechanisms behind the two cases are very different, and the interference comes from two different mechanisms.

V. INVESTIGATING THE OUTPUT OF THE NONLINEAR SCATTERER

Having performed the interaction with a nonlinear two-level system, we may be interested in the output quantum state of the scattered light, for instance, through the interferometer depicted in Fig. 3(b). To investigate this interference of the multimode output after scattering on the two-level system, we use a similar approach as above. We place a delay cavity in each output path of the atomic decay channels and choose the delay cavity along the top path to pick up the radiation in a mode $w_1(t)$, while the delay cavity in the lower path picks up the mode $w_2(t)$. We can pick $w_1(t)$ and $w_2(t)$ as we desire, for instance, the modes that carry the highest average number of excitations. A method for finding these modes is described in [22,23]. If we wish to investigate the multimode nature of the output field, we may pick up more modes using a number of delay cavities, but here we restrict ourselves to computing the output only in a single mode and treat all other modes as loss. Truncating the Hilbert space to only the relevant mode in question will yield the same results as if only the relevant mode is physically extracted for observation, e.g., by the quantum pulse gate [29].

For this example, we will analyze the output field from the nonlinear scatterer in the same mode as the input modes. We therefore choose the delay cavities to pick up $w_1(t) = v_1(t)$ and $w_2(t) = v_2(t)$, respectively. We can then reemit them in the same shape, but with a new relative delay τ' , if we desire to investigate their further propagation along another network, but we will not consider that here.

Introducing the two delay cavities after the atom will again increase the Hilbert space dimensionality, but we will employ an interaction picture, which reduces the size of the Hilbert space and makes the problem tractable. The approach is very similar to the one in the preceding section, and the derivation is deferred to Appendix A.

We denote the operators of the delay cavities in the second interferometer by $\hat{d}_i$, where $\hat{d}_1$ is for the cavity in the long path and $\hat{d}_2$ for the short path. We also rename $\hat{c}_+ = \hat{c}_1$ and $\hat{c}_- = \hat{c}_2$ in the following to ease the notation. In the interaction picture with respect to the linear transfer between all these cavity modes, the operators are given by a time-dependent rotation

of the initial operators at $t = 0$,

$$\hat{c}_i(t) = \cos \theta_i(t) \hat{c}_i(0) - \sin \theta_i(t) \hat{d}_i(0), \quad (29)$$

$$\hat{d}_i(t) = \cos \theta_i(t) \hat{d}_i(0) + \sin \theta_i(t) \hat{c}_i(0), \quad (30)$$

where the rotation angle is given by

$$\sin^2 \theta_i(t) = \int_0^t dt' |v_i(t')|^2. \quad (31)$$

The full Hamiltonian for this system in the interaction picture is $\hat{H}_I = \hat{H}_{\text{sys}} + \hat{H}_{\text{int}}$, where $\hat{H}_{\text{sys}} = \Delta \hat{\sigma}_+ \hat{\sigma}_-$ and

$$\hat{H}_{\text{int}} = i\sqrt{\frac{\gamma}{2}} \sum_{i=1}^2 v_i^*(t) \{ \hat{c}_i^\dagger \hat{\sigma}_- + \cot[2\theta_i(t)] \hat{d}_i^\dagger \hat{\sigma}_- \} + \text{H.c.}, \quad (32)$$

while dissipation operators in the interaction picture are similarly given by

$$\begin{aligned} \hat{L}_{1,I} &= \sqrt{2} \{ -v_1(t) \csc[2\theta_1(t)] \hat{d}_1 - v_2(t) \csc[2\theta_2(t)] \hat{d}_2 \} \\ &\quad + \sqrt{\gamma} \hat{\sigma}_-, \\ \hat{L}_{2,I} &= \sqrt{2} \{ v_1(t) \csc[2\theta_1(t)] \hat{d}_1 - v_2(t) \csc[2\theta_2(t)] \hat{d}_2 \}. \end{aligned} \quad (33)$$

We solve the master equation with this Hamiltonian and these dissipation operators to achieve the dynamics in the interaction picture. The atomic dynamics is the same in the Schrödinger picture and the interaction picture, but we notice that in the interaction picture, the $\hat{c}_i$ operators gradually rotate into the $\hat{d}_i$ operators, so the final population in the $\hat{c}_i$ oscillators in the interaction picture corresponds to a population in the $\hat{d}_i$ oscillators in the Schrödinger picture. Since the dynamics concerning the linear transfer of excitation between the cavities is handled by the interaction picture, the introduction of the two final delay cavities has simplified the dynamics, and this system is actually faster to compute numerically than the one without the final delay cavities [24]. This seems counterintuitive but originates from the fact that the states in the cavities no longer explore the full Hilbert space, but only a few states in the vicinity of the original photon number in the cavity at $t = 0$, since the two-level system can only handle a few excitations during the interaction. It is thus possible to treat large-photon-number states by truncating the Hilbert space to the few most relevant Fock states in the most populated modes in the interaction picture [24].

The integrated intensity of the interference of the two delayed output pulses on a second interferometer for $\tau' = 0$ is shown in Fig. 4 as a function of detuning. The intensity of the output field displays a behavior similar to the classical Ramsey effect. At zero detuning, the atom interacts strongly with the incoming pulses and the intensity in the pulse mode is reduced by one photon on average; while varying the detuning, a Ramsey-like interference pattern emerges.

VI. CONCLUSION AND OUTLOOK

We have presented a theoretical method to handle pulse propagation delays in optical networks. Two methods have been presented: one that can delay a single pulse an

arbitrary amount and one that can delay any number of pulses relative to other quantum channels. We presented examples of the theory and its application to evaluate the outcome of exciting a two-level atom by two entangled quantum pulses of radiation where we found a behavior similar to that of a classical drive, although the driving mechanisms are of very different physical nature. For a wide range of networks, our method provides a full quantum analysis, where there may be no agreement with classical results and where quantum features are deliberately strived for. Further exploration of the method may deal with dispersive propagation, described by a modified shape of the delayed pulses, sensing with interferometers, manipulation with time bin qubits, and studying time-delayed interactions between quantum emitters.

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DATA AVAILABILITY

The data that support the findings of this article are openly available [28].

APPENDIX A: DERIVATION OF THE INTERACTION PICTURE HAMILTONIAN

In this Appendix the interaction picture Hamiltonian and dissipation operators for the setup in Sec. V are derived, under the assumption that $w_i(t) = v_i(t)$. We take a starting point in the Hamiltonian and dissipation operators for an atom irradiated by a pulse which has traversed the Mach-Zehnder interferometer with different path lengths in Sec. III, which is given by

$$\hat{H}_1 = \Delta\hat{\sigma}_+\hat{\sigma}_- + \frac{i}{2}\sqrt{\frac{\gamma}{2}}[g_{\text{out},v_1}(t)\hat{c}_1^\dagger\hat{\sigma}_- + g_{\text{out},v_2}(t)\hat{c}_2^\dagger\hat{\sigma}_- - \text{H.c.}], \quad (\text{A1})$$

where we have introduced the operators $\hat{c}_1 = \frac{1}{\sqrt{2}}(\hat{a}_u + \hat{b}_2)$ and $\hat{c}_2 = \frac{1}{\sqrt{2}}(\hat{a}_u - \hat{b}_2)$, which also simplifies the dissipation operators

$$\hat{L}_1 = g_{\text{out},v_1}^*(t)\hat{c}_1 + \sqrt{\frac{\gamma}{2}}\hat{\sigma}_-, \quad (\text{A2})$$

$$\hat{L}_2 = g_{\text{out},v_2}^*(t)\hat{c}_2 + \sqrt{\frac{\gamma}{2}}\hat{\sigma}_-. \quad (\text{A3})$$

Appending delay cavities to the two output ports that pick up the radiation in the $v_1(t)$ and $v_2(t)$ modes results in the

addition of the following terms to the Hamiltonian:

$$\begin{aligned} \hat{H}_2 = \hat{H}_1 + \frac{i}{2} \left(\sqrt{\frac{\gamma}{2}} [g_{\text{in},v_1}^*(t)\hat{\sigma}_+\hat{d}_1 + g_{\text{in},v_2}^*(t)\hat{\sigma}_+\hat{d}_2] \right. \\ \left. + g_{\text{out},v_1}(t)g_{\text{in},v_1}^*(t)\hat{c}_1^\dagger\hat{d}_1 \right. \\ \left. + g_{\text{out},v_2}(t)g_{\text{in},v_2}^*(t)\hat{c}_2^\dagger\hat{d}_2 - \text{H.c.} \right). \end{aligned} \quad (\text{A4})$$

The dissipation operators are

$$\begin{aligned} \hat{L}_1 = \frac{1}{\sqrt{2}} [g_{\text{in},v_1}^*(t)\hat{d}_1 + g_{\text{in},v_2}^*(t)\hat{d}_2] \\ \frac{1}{\sqrt{2}} [g_{\text{out},v_1}(t)\hat{c}_1 + g_{\text{out},v_2}(t)\hat{c}_2] + \sqrt{\gamma}\hat{\sigma}_-, \end{aligned} \quad (\text{A5})$$

$$\begin{aligned} \hat{L}_2 = \frac{1}{\sqrt{2}} [g_{\text{in},v_2}^*(t)\hat{d}_2 - g_{\text{in},v_1}^*(t)\hat{d}_1] \\ + \frac{1}{\sqrt{2}} [g_{\text{out},v_2}(t)\hat{c}_2 - g_{\text{out},v_1}(t)\hat{c}_1]. \end{aligned} \quad (\text{A6})$$

Now we go to the interaction picture with respect to the cavity-cavity part of the Hamiltonian

$$\begin{aligned} \hat{H}_0 = \frac{i}{2} [g_{\text{out},v_1}(t)g_{\text{in},v_1}^*(t)\hat{c}_1^\dagger\hat{d}_1 \\ + g_{\text{out},v_2}(t)g_{\text{in},v_2}^*(t)\hat{c}_2^\dagger\hat{d}_2 - \text{H.c.}]. \end{aligned} \quad (\text{A7})$$

This leads to the equations of motion for the interaction picture operators

$$\frac{d}{dt}\hat{c}_i = \frac{1}{2}g_{\text{out},v_i}(t)g_{\text{in},v_i}^*(t)\hat{d}_i, \quad (\text{A8})$$

$$\frac{d}{dt}\hat{d}_i = -\frac{1}{2}g_{\text{in},v_i}(t)g_{\text{out},v_i}^*(t)\hat{c}_i \quad (\text{A9})$$

for $i = 1, 2$. The solution to these differential equations is the same as we saw in the main text,

$$\hat{c}_i(t) = \cos\theta_i(t)\hat{c}_i(0) - \sin\theta_i(t)\hat{d}_i(0), \quad (\text{A10})$$

$$\hat{d}_i(t) = \cos\theta_i(t)\hat{d}_i(0) + \sin\theta_i(t)\hat{c}_i(0), \quad (\text{A11})$$

where $\theta_i(t)$ is defined by

$$\sin^2\theta_i(t) = \int_0^t dt' |v_i(t')|^2. \quad (\text{A12})$$

This transformation gives that the Hamiltonian in the interaction picture is

$$\begin{aligned} \hat{H}_I = \Delta\hat{\sigma}_+\hat{\sigma}_- + i\sqrt{\frac{\gamma}{2}} \sum_{i=1}^2 v_i^*(t) \{\hat{c}_i^\dagger\hat{\sigma}_- \\ + \cot[2\theta_i(t)]\hat{d}_i^\dagger\hat{\sigma}_- - \text{H.c.}\} \end{aligned} \quad (\text{A13})$$

and the dissipation operators become

$$\begin{aligned} \hat{L}_{1,I} = \sqrt{2} \{-v_1(t) \csc[2\theta_1(t)]\hat{d}_1 - v_2(t) \csc[2\theta_2(t)]\hat{d}_2\} \\ + \sqrt{\gamma}\hat{\sigma}_-, \end{aligned} \quad (\text{A14})$$

$$\hat{L}_{2,I} = \sqrt{2} \{v_1(t) \csc[2\theta_1(t)]\hat{d}_1 - v_2(t) \csc[2\theta_2(t)]\hat{d}_2\}. \quad (\text{A15})$$

APPENDIX B: CLASSICAL INTERACTION

In this Appendix we present the details of the calculation for the classical drive of the atom. For the quantum drive, we used a Fock state with n photons, which were split upon a beam splitter into an early and a late pulse. In the classical calculation, we assume two classical drives with an average number of photons $\alpha = \sqrt{n/2}$, so the average numbers of photons in the pulses are the same in the classical and quantum drives. The Hamiltonian for the classical drive is

$$\hat{H}_c = \hat{H}_{\text{sys}} + i\sqrt{\frac{\gamma}{2}} \sum_{i=1}^2 v_i^*(t) \alpha^* \hat{\sigma}_- + \text{H.c.}, \quad (\text{B1})$$

where $\hat{H}_{\text{sys}}$ is defined similarly to the one in the main text and the pulse parameters for $v_1(t)$ and $v_2(t)$ are also the same. We also include a dissipation channel similar to the quantum case

$$\hat{L}_c = \sqrt{\gamma} \hat{\sigma}_-. \quad (\text{B2})$$

The master equation is solved numerically with this Hamiltonian and dissipation term to yield the atomic state population as a function of time and detuning.

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